IMAGING COURSE · LESSON 10

The neck by spaces

The neck looks like a crowded jumble on CT until you see it as a set of tubes and compartments wrapped by the three layers of the deep cervical fascia. Each compartment, or **space**, contains a limited set of normal structures, and nearly every lesion in it comes from one of them. Harnsberger's insight, the backbone of *Diagnostic Imaging: Head and Neck*, is that if you can name the space a mass arises in, the differential diagnosis shrinks to a few lines. This lesson teaches the map and the one trick that tells you where a deep mass began.

AT A GLANCE

- Name the **space** first: its contents give the differential.
- Suprahyoid: four spaces (**mucosal, masticator, parotid, carotid**) around the **parapharyngeal fat**; retropharyngeal and perivertebral behind.
- Fat pushed **laterally**: mucosal; **posteriorly**: masticator; **medially**: deep parotid; **anteriorly**: carotid.
- Carotid body paraganglioma **splays** the ICA and ECA; vagal tumours push the ICA **forwards**.
- Cellulitis, phlegmon and abscess differ on contrast CT; only an abscess is drained. The danger space reaches the mediastinum.

1 CHAPTER ONE The fascia and the spaces

A map of the neck.

1.1 Three layers, many spaces

The **superficial layer** of the deep cervical fascia invests the neck: it surrounds the masticator and parotid spaces above the hyoid and the sternocleidomastoid, trapezius and strap muscles below. The **middle layer** lines the deep surface of the pharyngeal mucosa above and wraps the visceral space (larynx, trachea, thyroid, oesophagus) below. The **deep layer** surrounds the perivertebral space, and a slip of it, the **alar fascia**, separates the retropharyngeal space from the danger space behind it. All three contribute to the **carotid sheath**.

1.2 The suprahyoid neck

Above the hyoid, a pair of fat-filled **parapharyngeal spaces** sits at the centre, surrounded by four key spaces: the **pharyngeal mucosal space** medially, the **masticator space** in front, the **parotid space** laterally and the **carotid space** behind. In the midline behind the pharynx are the **retropharyngeal space** and the **perivertebral space**.

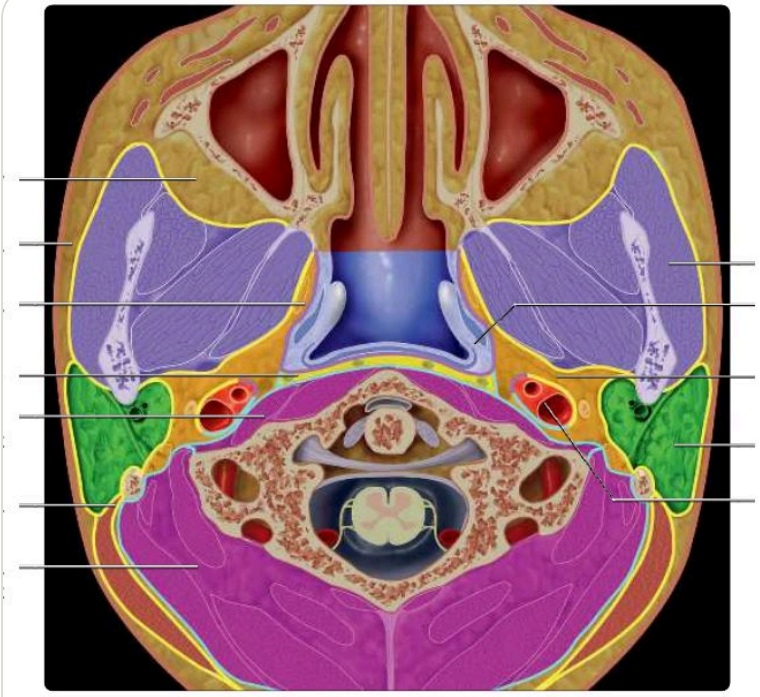


FIGURE 1 The spaces of the suprahyoid neck at the level of the nasopharynx, colour-coded: pharyngeal mucosal space (blue), masticator spaces (purple), parotid spaces (green), carotid spaces with the artery and vein (red), the fat of the parapharyngeal spaces (yellow-orange) in the middle of them, and the perivertebral space (magenta) behind. The fascial layers are drawn as coloured lines.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 6

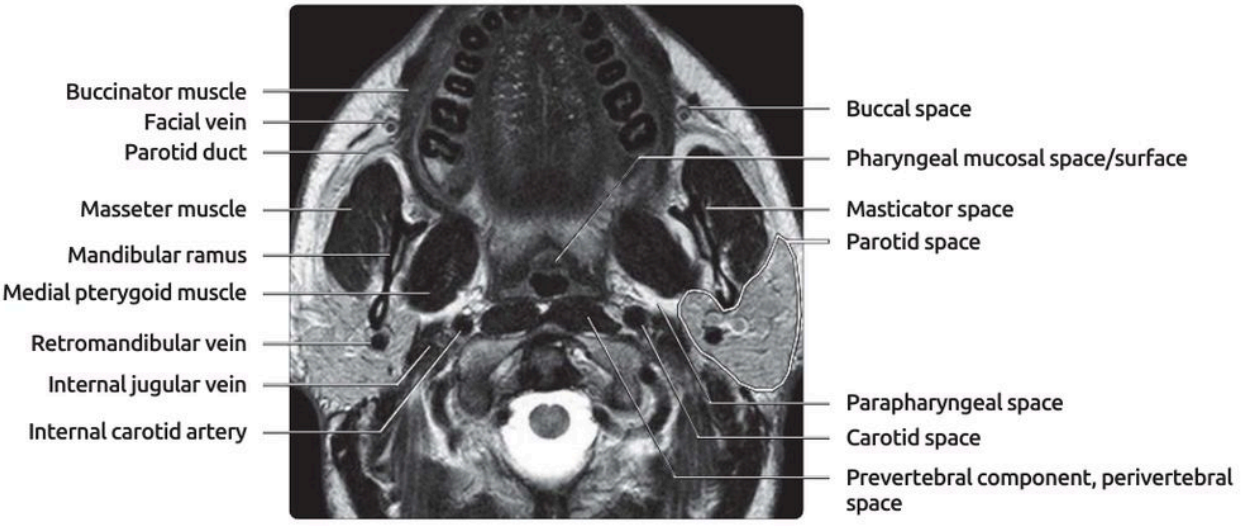


FIGURE 2 The same spaces on axial T2 MRI: the pharyngeal mucosal space, masticator space, parotid space (outlined) and carotid space arranged around the parapharyngeal fat, with the prevertebral muscles behind.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 284

1.3 The infrahyoid neck

Below the hyoid, the **visceral space** in the centre holds the larynx and hypopharynx, the trachea and oesophagus, the thyroid and parathyroid glands, and the recurrent laryngeal nerves. On either side are the **carotid spaces**, behind them the **posterior cervical spaces** (fat, the accessory nerve and level V nodes), and in the midline behind are the retropharyngeal and perivertebral spaces. The **anterior cervical space** is the fat between the carotid space and the visceral space.

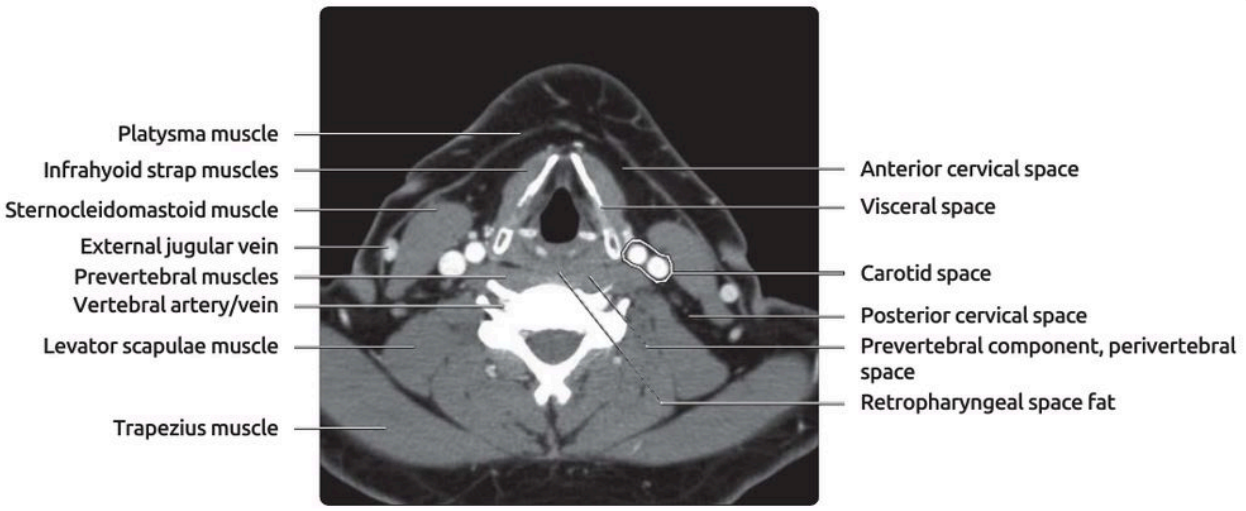


FIGURE 3 Axial contrast CT at the glottis: the visceral space with the larynx and strap muscles, the carotid spaces behind it on either side, the posterior cervical spaces and the perivertebral space.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 279

CHAPTER TWO

2 What each space contains

And therefore what arises there.

THE SPACES, THEIR CONTENTS AND THEIR COMMON LESIONS

SPACE	NORMAL CONTENTS	COMMON LESIONS
Pharyngeal mucosal	Mucosa, lymphoid tissue (adenoids, tonsils), minor salivary glands, constrictor muscles	Squamous cell carcinoma, lymphoma, tonsillar abscess, minor salivary tumour
Parapharyngeal	Fat, occasional minor salivary rests	Rare primary lesions; mostly displaced by its neighbours. Pleomorphic adenoma of salivary rests, lipoma
Masticator	Mandibular ramus and body, muscles of mastication, V3, pterygoid venous plexus	Odontogenic abscess, sarcoma, perineural tumour on V3, benign masseter hypertrophy
Parotid	Parotid gland, facial nerve, intraparotid nodes, retromandibular vein	Pleomorphic adenoma, Warthin tumour, carcinoma, nodes

SPACE	NORMAL CONTENTS	COMMON LESIONS
Carotid	Internal carotid, internal jugular vein, IX to XII (only X below the hyoid), sympathetic chain	Paraganglioma, schwannoma, thrombosis, nodes alongside
Retropharyngeal	Fat; lateral (Rouvière) and medial retropharyngeal nodes above the hyoid	Nodal metastasis, suppurative node, abscess, oedema
Perivertebral	Vertebrae, prevertebral and scalene muscles, brachial plexus, vertebral artery	Discitis and osteomyelitis, metastasis, chordoma, brachial plexus schwannoma
Visceral	Larynx, hypopharynx, trachea, oesophagus, thyroid, parathyroids	Carcinoma of each, thyroid and parathyroid lesions
Posterior cervical	Fat, accessory nerve, level V nodes	Nodes, lymphatic malformation, lipoma, schwannoma

CHAPTER THREE

3 Where did it start?

The parapharyngeal fat tells you.

3.1 The displacement rule

The parapharyngeal space is mostly fat and rarely the origin of disease, so it acts as a marker. A mass growing in one of the four surrounding spaces pushes the fat away from itself:

- a **pharyngeal mucosal** mass pushes the fat **laterally**;
- a **masticator** space mass pushes it **posteriorly** (and medially);
- a **deep lobe parotid** mass pushes it **medially** (and widens the stylomandibular gap);
- a **carotid space** mass pushes it **anteriorly**, and lifts the styloid process forwards;
- a lateral **retropharyngeal** mass pushes it anteriorly without displacing the styloid.

A mass completely surrounded by fat arises in the parapharyngeal space itself, usually from salivary rests.

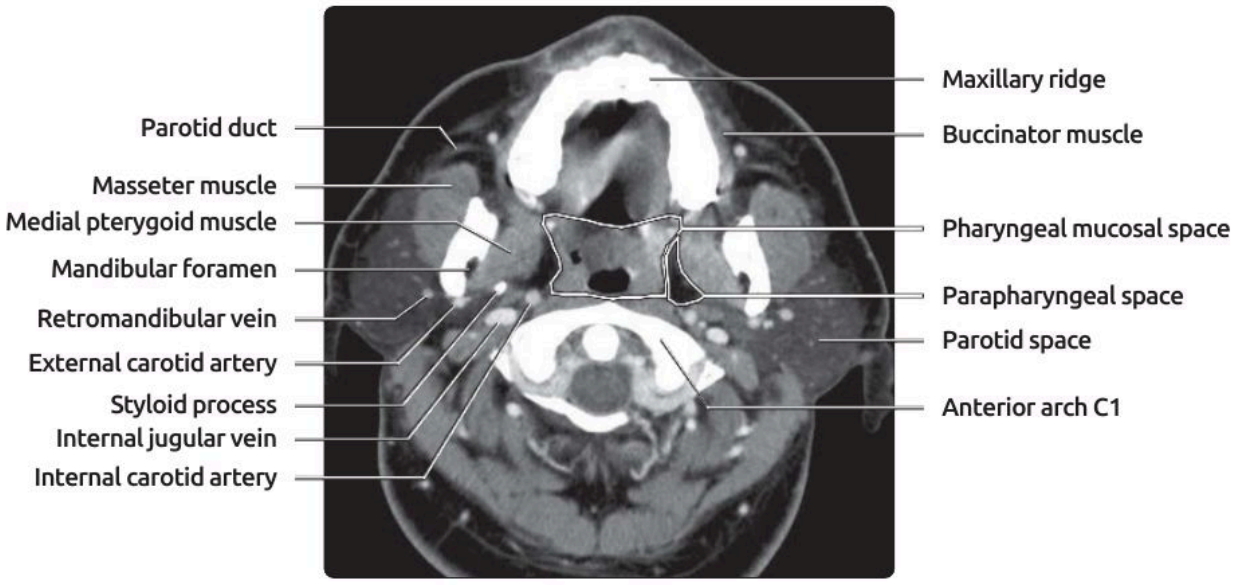


FIGURE 4 Axial contrast CT at the level of the maxillary ridge: the pharyngeal mucosal space outlined between the paired parapharyngeal fat pads, with the masticator spaces in front and the carotid spaces behind.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 277



FIGURE 5 A true parapharyngeal space pleomorphic adenoma. Left: CT, a well-defined mass completely surrounded by parapharyngeal fat. Right: T1 MRI, the mass separate from the deep lobe of the parotid, with a rim of fat all round.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 14

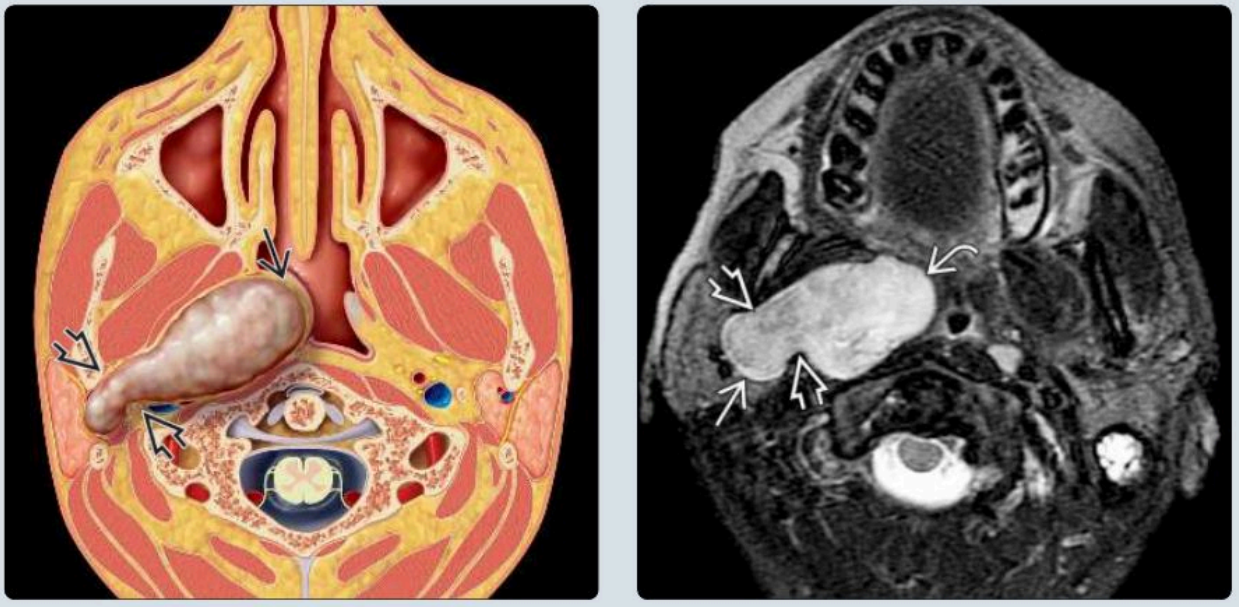


FIGURE 6 A deep lobe parotid pleomorphic adenoma. Left: drawing of a pear-shaped tumour pushing the parapharyngeal fat medially. Right: T2, the bright tumour passing through a widened stylomandibular notch and displacing the fat and the tonsil medially.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 88

3.2 The carotid space and the vessels

In the carotid space, the position of the vessels refines the diagnosis. A **carotid body paraganglioma** sits in the bifurcation and **splays** the internal and external carotid arteries apart. A **vagal** paraganglioma or schwannoma, higher up, pushes the internal carotid **anteriorly** (and medially). A **sympathetic** schwannoma, medial to the vessels, separates neither. Paragangliomas enhance intensely and show flow voids ("salt and pepper"); schwannomas enhance less and may be cystic.

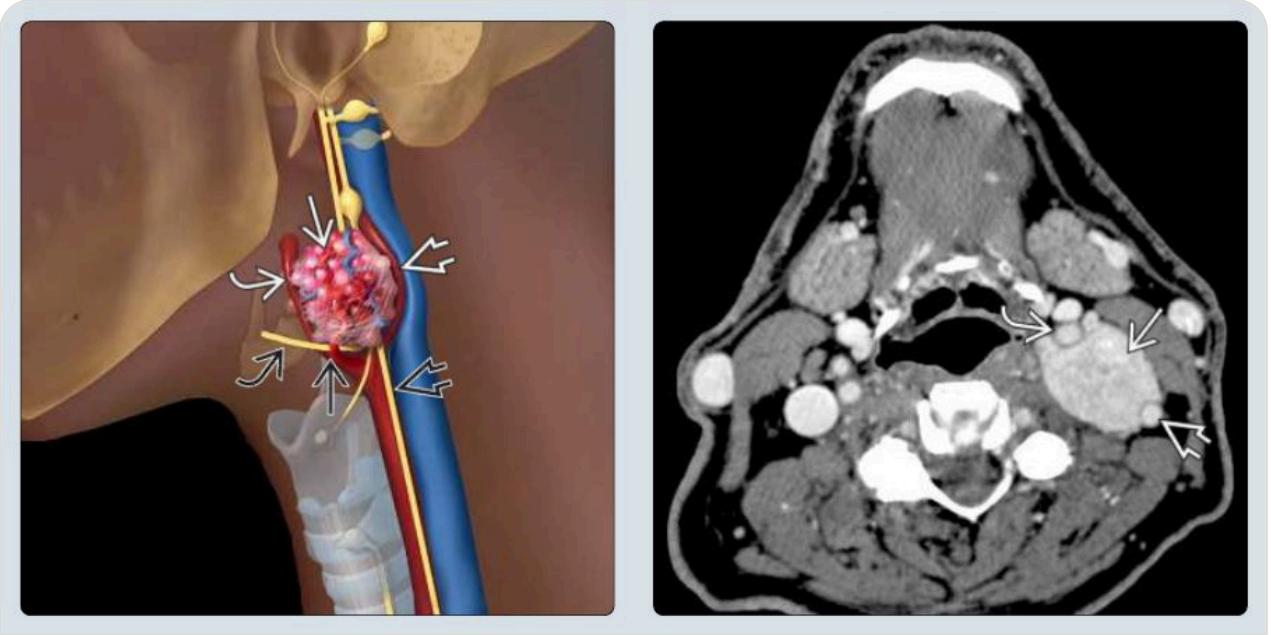


FIGURE 7 Carotid body paraganglioma. Left: drawing of the tumour in the bifurcation, splaying the internal and external carotids. Right: contrast CT, an avidly enhancing mass sitting in the notch between the two arteries.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 140

4 CHAPTER FOUR Spaces as pathways

How disease spreads.

4.1 Infection

The spaces also explain how infection travels. A **peritonsillar abscess** can rupture through the superior constrictor into the parapharyngeal fat and on to the masticator or carotid spaces. Dental infection enters the masticator, submandibular and sublingual spaces. The **retropharyngeal space** extends down to about T1–T6, and the **danger space** behind it to the diaphragm, so infection there threatens the mediastinum. A suppurative retropharyngeal node in a child (a rim-enhancing node, often lateral) is not the same as a true retropharyngeal abscess (fluid distending the space from side to side, often with mass effect on the airway).

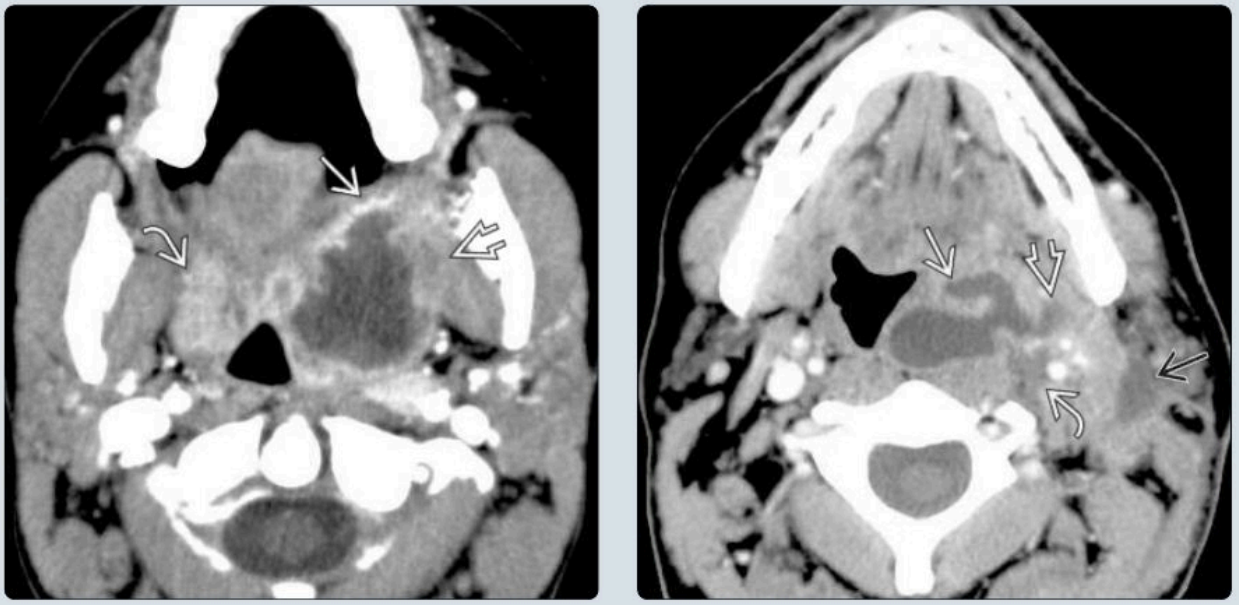


FIGURE 8 Tonsillar abscess complications. Left: a tonsillar abscess extending through the capsule into the buccal space and the medial pterygoid. Right: a large abscess that has ruptured into the carotid space and the submandibular space.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28

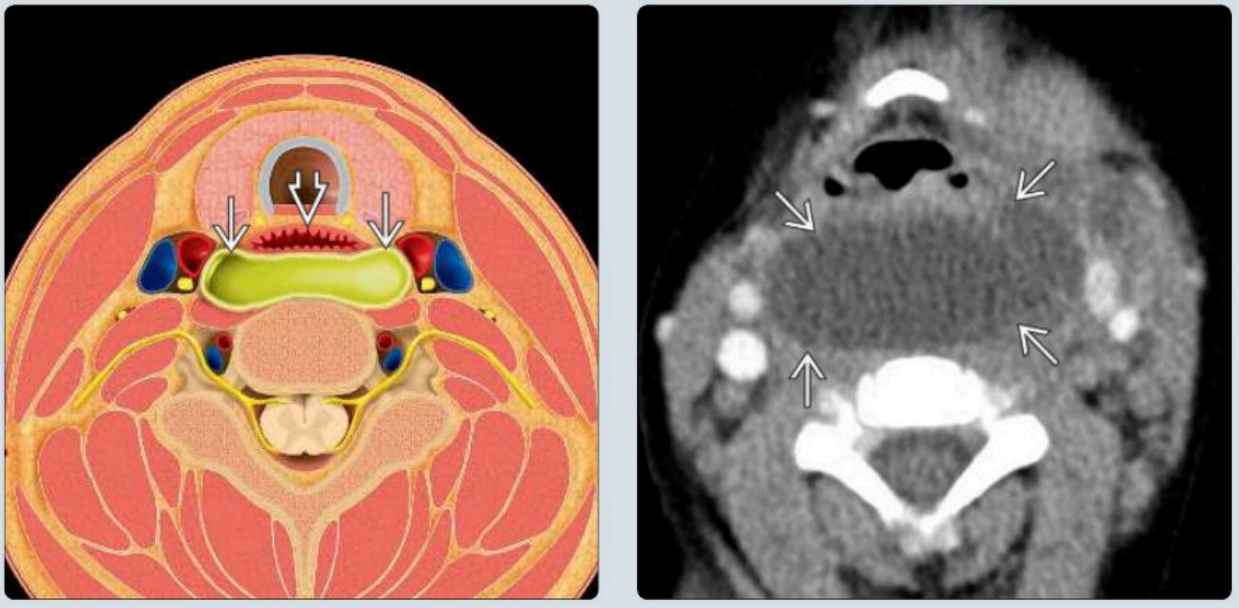


FIGURE 9 Retropharyngeal abscess. Left: drawing of the collection behind the pharynx. Right: contrast CT in an infant, a low-density collection distending the retropharyngeal space and pushing the pharynx forwards.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 168

4.2 Abscess, phlegmon or cellulitis?

On contrast CT, **cellulitis** is stranding of the fat; **phlegmon** is ill-defined, enhancing soft tissue without a fluid centre; an **abscess** is a fluid collection with a well-formed enhancing rim. Only a mature abscess is drained. Look also for the airway, the internal jugular vein (thrombosis, Lemierre syndrome), the carotid, the mediastinum and the cause (tooth, stone, tonsil, foreign body).

PLACING A DEEP NECK MASS

1. Above or below the hyoid
   - → **suprahyoid** — find the parapharyngeal fat
   - → **infrahyoid** — visceral, carotid, posterior cervical
2. Which way is the fat pushed
   - → **laterally** — mucosal
   - → **posteriorly** — masticator
   - → **medially** — deep parotid
   - → **anteriorly** — carotid or retropharyngeal
   - → **surrounded** — parapharyngeal itself
3. What normally lives there
   - → match to the contents table
4. Look at the vessels
   - → **splayed** — carotid body

→ **ICA pushed forwards**
vagal

→ **stylomandibular gap widened**
deep parotid

5 **Character**

→ cyst, fat, enhancement, flow voids, bone

6 **Differential**

→ two to four diagnoses by space and character

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The three layers** of deep cervical fascia and what they wrap.
- **The suprahyoid spaces** around the parapharyngeal fat, and the **infrahyoid spaces**.
- **The displacement rule** for the parapharyngeal fat.
- **Contents of each space** and the lesions that arise there.
- **Carotid space masses:** carotid body versus vagal paraganglioma versus schwannoma.
- **Danger space** and mediastinal spread; abscess versus phlegmon.

5.2 Common pitfalls

COMMON PITFALLS

- Describing a deep neck mass without naming its space.

- Calling a deep lobe parotid tumour "parapharyngeal" when the fat is pushed medially.
- Draining a phlegmon.
- Calling a suppurative retropharyngeal node a retropharyngeal abscess.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), suprahyoid and infrahyoid neck overview, Imaging Anatomy: Head and Neck (Chapman and Koontz), neck spaces chapters, ESR Modern Radiology, Head and Neck Imaging. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.